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Machine learning

Data — Training / Test

Supervised	unsupervised.
input - output given	unlabeled data.
- Classification	- Clustering
- regression	- dimensionality reduction

Supervised

→ input x to predict an output y

x may be

↳ feature vector

↳ sequences

↳ Matrices

↳ graphs.

Mathematically $f: x \rightarrow y$

Define a loss function that quantifies the error

. Optimize model parameters based off of it.

Two steps:

Training : train on data to ~~[illegible]~~ learn a model.
Testing : use a model on unseen data to assess accuracy

$$\text{Accuracy} = \frac{\text{No. of correct classifications}}{\text{total test cases}}.$$

Regression

x : input variables / features
y : output variables / target

We'll see simple univariate Linear regression.

- one independent and one dependent variable
- related by a straight line.

simple linear regression model:

$$y = b_0 + b_1 x + \varepsilon.$$

where b_0 and b_1 are parameters of the model
(the values we need the model to learn)

ε is the error term

simple linear regression equation:

$$\boxed{\hat{y} = b_0 + b_1 x}$$

where b_0 = y-intercept
b_1 = slope of the regression line
$\hat{y}$ = expected value.

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Loss-function - Mean squared error (MSE)

- Measures how far is the made prediction from the actual value.

$$MSE = \frac{\sum(y_i - \hat{y}_i)^2}{n}$$

where y_i = observed value
$\hat{y}_i$ = estimate value.

Least squares Method

Least squares criterion:

$$\min \sum(y_i - \hat{y}_i)^2$$

where y_i = observed value
$\hat{y}_i$ = estimated value.

Slope:

$$\boxed{b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}}$$

y-intercept

$$\boxed{b_0 = \bar{y} - b_1 \bar{x}}$$

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Variability

High: data points are further apart
Low: data points are close together.

measured in:

range : $\max(X) - \min(X)$.

variance: $\frac{1}{n-1}\left(\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right)$

standard deviation: $\sqrt{\text{variance}}$

Coefficient of determination

Total sum of squares — regression sum of squares — error sum of squares.

$$SST = SSR + SSE$$

$$\Sigma(y_i - \bar{y})^2 = \Sigma(\hat{y}_i - \bar{y})^2 + \Sigma(y_i - \hat{y}_i)^2.$$

$$\boxed{r^2 = \frac{SSR}{SST}} = \boxed{1 - \frac{SSE}{SST}}$$

$r^2 = 0$ model explains none of variability

$r_2 = 1$ perfect fit

$r = 0.75$ 75% of variation in y is explained by x.

CamScanner

Sample Correlation Coefficent.

$$r_{xy} = (\text{sign of } b_1)(\sqrt{r_2})$$

$1 \leq r_{xy} \leq -1.$

$r_{xy} > 1$	Positive linear relationship
$r_{xy} < 1$	Negative linear relationship.
$r_{xy} = 0$	No linear relationship.
$r_{xy} = 1/-1$	strong linear relationship.